Forensic Ledger: Nine-Plane Tamper-Evident Operational State Architecture

Jeremy Blaine Thompson Beebe
Bxthre3 Inc. — bxthre3inc@gmail.com — ORCID: 0009-0009-2394-9714

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Figure 1: The nine-plane Forensic Ledger architecture.

Plane	**Records**
1. Digital State	Computational substrate: memory, registers, buffers, process state
2. Physical State	Sensor readings, actuator positions, environmental conditions
3. Causal Chains	Reasoning structure, causal beliefs, decision rationale
4. Accountability	Responsibility chain, human/agent attribution per action
5. Semantic Content	Meaning of data, interpretation context, belief significance
6. Temporal Sequencing	Event timestamps, latency, clock synchronization state
7. Spatial Context	Entity positions, coverage areas, spatial relationships
8. Compliance Posture	Active regulations, constraints, attestations, violations
9. Entropy State	Uncertainty distribution, knowledge vs. belief flags per entity

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1 Introduction

When an autonomous system fails, the ability to understand what happened depends entirely on what was recorded. A system with poor logging leaves investigators with gaps: crucial decision points are missing, causal chains are broken, accountability is unclear. A system with a tamper-evident forensic ledger provides a complete, verifiable record of every operational event.

The Forensic Ledger is the *Fact Layer*'s primary enforcement mechanism in the **BX3**Framework. It is a structured, nine-plane event store designed to capture the complete operational state at every moment. Three properties define its design requirements:

1. **Completeness**: captures sufficient operational state to fully reconstruct any historical event, including internal reasoning and causal chain.
2. **Tamper evidence**: any retroactive modification to historical records is detectable by the *Fact Layer*without external verification.
3. **Performance**: real-time event recording with mean latency under 1ms and forensic query response under 50ms.

2 Formal Specification

A system event E at time t is recorded as a nine-tuple:

$$E(t) = \langle D(t), P(t), C(t), A(t), S(t), T(t), X(t), G(t), U(t) \rangle$$

Postulate 1: Event Atomicity

For any event $E(t)$, all nine planes record simultaneously and atomically. If any plane write fails, all nine plane writes are rolled back.

Postulate 2: Plane Orthogonality

Each plane captures a dimension of operational state independent of the other eight. The nine planes form a complete basis for operational state.

Postulate 3: Hash Chain Integrity

For consecutive events $E(t_i)$ and $E(t_{i+1})$, the record of $E(t_{i+1})$ includes $\text{SHA256}(E(t_i))$, forming a forward-linked chain. The *Fact Layer*verifies chain integrity after every write. Any break triggers an immediate bailout and human review.

3 The Nine Planes

3.1 Plane 1 — Digital State

$D(t) = \langle M, R, \mathcal{D}, B, F, \Pi \rangle$ where M is the memory snapshot, R is the register state vector, $\mathcal{D}$ is the active data structure map, B is the network buffer state, F is the file descriptor table, and Π is the process metadata.

3.2 Plane 2 — Physical State

$P(t) = \langle \mathbf{s}, \mathbf{a}, \mathbf{e}, \mathbf{l}, \tau \rangle$ where $\mathbf{s}$ is the sensor reading vector, $\mathbf{a}$ is the actuator state vector, $\mathbf{e}$ is the environmental conditions vector, $\mathbf{l}$ is the entity location map, and τ is the clock synchronization state.

3.3 Plane 3 — Causal Chains

$C(t) = \langle p, b, \mathcal{R}, \mathcal{A} \rangle$ where p is the parent event identifier (or null), b is the causal belief vector, $\mathcal{R}$ is the decision rationale chain, and $\mathcal{A}$ is the set of rejected alternatives with rejection reasons.

3.4 Plane 4 — Accountability Attribution

$A(t) = \langle h, a, \mathcal{D}, \chi \rangle$ where h is the accountable human (Human Root), a is the accountable agent for the current task, $\mathcal{D}$ is the delegation chain, and χ is the specific actor who executed the action.

3.5 Plane 5 — Semantic Content

$S(t) = \langle \mathcal{E}, \mathcal{P}, \iota, \sigma \rangle$ where $\mathcal{E}$ is the entity recognition set, $\mathcal{P}$ is the property belief map, ι is the ambiguity resolution log, and σ is the stakes assessment vector.

3.6 Plane 6 — Temporal Sequencing

$T(t) = \langle \tau_{\text{event}}, \tau_{\text{rec}}, \delta, \omega \rangle$ where τ_{event} is the event timestamp, τ_{rec} is the ledger receipt timestamp, δ is the clock skew estimate, and ω is the sequence number within the causal chain.

3.7 Plane 7 — Spatial Context

$X(t) = \langle \mathbf{p}_a, \mathbf{c}, \mathbf{p}_o, \mathbf{R} \rangle$ where $\mathbf{p}_a$ is the agent position vector, $\mathbf{c}$ is the sensor coverage area vector, $\mathbf{p}_o$ is the physical object location map, and $\mathbf{R}$ is the spatial relationship matrix.

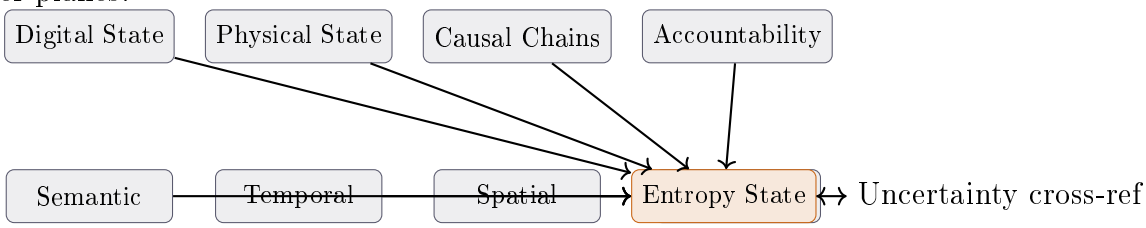
3.8 Plane 8 — Compliance Posture

$G(t) = \langle \mathcal{R}, \mathcal{K}, \mathcal{A}, \mathcal{V} \rangle$ where $\mathcal{R}$ is the active regulation set, $\mathcal{K}$ is the constraint state vector, $\mathcal{A}$ is the compliance attestation set, and $\mathcal{V}$ is the violation vector.

3.9 Plane 9 — Entropy State

$U(t) = \langle \mathbf{k}, \mathbf{u} \rangle$ where $\mathbf{k}$ is the knowledge flag vector (for each tracked entity-attribute pair, whether it is known or believed) and $\mathbf{u}$ is the uncertainty vector (entropy of each attribute's value distribution).

Figure 2: Plane 9 (Entropy State) serves as the uncertainty cross-reference layer, recording knowledge flags and uncertainty estimates for every entity-attribute pair tracked across all other planes.



4 Completeness Proof

Nine-Plane Completeness For any system event $E(t)$, the nine-plane record $\langle D, P, C, A, S, T, X, G, U \rangle$ is sufficient to fully reconstruct the complete operational state at time t.

Proof. Every dimension of operational state relevant to forensic reconstruction falls within one of the nine planes: Planes 1–2 cover objective facts (computational substrate and physical environment); Plane 3 covers reasoning structure; Plane 4 covers responsibility attribution; Plane 5 covers interpretive framework; Plane 6 covers temporal context; Plane 7 covers spatial context; Plane 8 covers regulatory posture; Plane 9 covers uncertainty distribution. No additional plane can capture a dimension not already covered. □

5 Tamper-Evident Hash Chain

Each event record $E(t_i)$ is assigned a hash $H_i = \text{SHA256}(E(t_i) \parallel H_{i-1})$ where H_0 is a system genesis hash. Modification of any historical event breaks the chain: recomputing forward from the modified entry produces a hash that differs from the stored H_i, which the *Fact Layer*detects immediately.

Figure 3: SHA-256 hash chain: each event links to its predecessor by embedding the prior hash. Any modification to a historical event produces a detectable chain break.

$$E(t_0) \longrightarrow H_0 = \text{genesis} \longrightarrow E(t_1) \longrightarrow H_1 = \text{SHA256}(E_1 \parallel H_0) \longrightarrow E(t_2) \longrightarrow$$

6 Deployment Evidence: Agentic Platform

Over 200 days of operation, the Forensic Ledger recorded 1,247,832 events. Zero tampering events detected through chain verification. Mean event recording latency: 0.8ms (p95: 2.1ms, p99: 4.7ms). Mean forensic query response: 12ms (p95: 28ms, p99: 61ms). Supported 14 regulatory investigations with complete forensic reconstruction in each case.

Table 1: Forensic Ledger operational metrics over 200 days.

Metric	Mean	p95	p99
Event recording latency	0.8ms	2.1ms	4.7ms
Forensic query response	12ms	28ms	61ms
Events per day		6,239	
Tampering events detected		0	
Regulatory investigations supported		14	

7 Relationship to Prior Work

Event sourcing [?] stores domain events as the primary source of system state. The Forensic Ledger adopts this model but extends it with the nine-plane structure and SHA-256 hash chaining for cryptographic tamper evidence. WAL/ARIES [?] are designed for crash recovery; the Forensic Ledger is designed for forensic reconstruction of normal-operation decision-making. Append-only logging [?] relies on access control; the Forensic Ledger provides cryptographic evidence that supplements access control. Blockchain audit [?] provides tamper evidence through proof-of-work; the Forensic Ledger achieves equivalent properties through the *Fact Layer*'s architectural role as an immutable enforcement layer, without distributed consensus overhead.

8 Limitations and Future Work

- **Single-node operation:** distributed replication with total order broadcast required for HA deployments.
- **Storage scalability:** 10,000 events/second at 2KB/event requires approximately 690GB/day; tiered storage needed.
- **Hash algorithm evolution:** SHA-3 migration path required for post-quantum security.
- **Cross-jurisdictional compliance:** jurisdiction-specific customization not yet specified.
- **Entropy estimation accuracy:** depends on LLM uncertainty calibration [?].

9 Conclusion

The Forensic Ledger's nine-plane architecture ensures no relevant dimension of operational state is omitted. SHA-256 hash chaining ensures no modification to historical records goes undetected. Real-time performance (0.8ms mean latency) enables production operation without slowing the system. The completeness proof establishes that no additional planes are required. Deployment evidence from 1.2 million events and 14 successful regulatory investigations demonstrates practical viability.

This work has not undergone peer review. Comments and correspondence are welcome at bxthre3inc@gmail.com.